

MAGADAN SOKOL, Russian FederationWMO: **259130**Lat: **59.911N**Lon: **150.720E**Elev: **175**StdP: **99.24**Time Zone: **11.00 (E11)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.0	-36.9	-42.7	0.1	-38.7	-40.1	0.1	-36.4	10.9	-6.6	9.6	-7.1	1.2	50	0.796

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	23.0	15.2	20.9	14.4	18.9	13.3	16.3	21.1	15.1	19.2	14.1	17.5	3.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.2	10.3	17.4	13.2	9.7	16.0	12.5	9.2	15.1	46.4	21.1	42.9	19.1	40.1	17.5	20.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	7.6	6.6	-41.8	26.6	2.3	2.1	-43.4	28.1	-44.8	29.3	-46.1	30.6	-47.8	32.1		
(5)			-41.7	18.5	2.2	1.2	-43.3	19.4	-44.6	20.1	-45.8	20.7	-47.4	21.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.0	-20.4	-19.6	-14.0	-5.6	3.4	9.9	13.8	12.8	6.9	-3.0	-13.5	-19.3
	DBStd	13.75	7.72	7.11	6.49	5.26	3.11	3.09	2.36	2.36	3.78	5.53	7.22	7.64
	HDD10.0	5352	942	828	744	468	205	39	1	5	104	404	704	907
	HDD18.3	8137	1201	1062	1003	718	462	252	143	172	342	663	954	1166
	CDD10.0	258	0	0	0	0	1	37	117	92	12	0	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	47	0	0	0	0	1	7	34	5	0	0	0	0
	CDH26.7	4	0	0	0	0	0	1	3	0	0	0	0	0
Wind	WSAvg	2.6	2.6	2.5	2.7	2.8	2.7	2.7	2.4	2.3	2.3	2.5	2.7	2.7
Precipitation	PrecAvg	534	14	12	17	26	32	49	69	103	75	61	52	24
	PrecMax	745	61	36	67	156	101	127	298	238	173	150	142	115
	PrecMin	316	0	1	0	2	2	2	3	32	9	10	1	1
	PrecStd	128	13	10	14	31	23	30	57	55	39	33	38	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.1	0.0	1.9	8.1	19.0	24.1	26.9	24.0	18.1	10.0	3.2	0.1
		MCWB	-1.9	-2.0	-1.2	2.9	10.1	14.2	17.3	16.4	11.9	5.2	1.3	-1.2
	2%	DB	-2.4	-2.7	0.0	5.1	14.9	21.2	23.9	21.9	15.8	7.5	1.8	-2.2
		MCWB	-3.8	-4.3	-2.4	1.0	8.2	13.3	16.3	14.9	11.0	4.4	0.3	-3.3
	5%	DB	-5.6	-5.3	-1.8	3.2	12.1	18.8	21.4	19.8	14.0	5.8	0.1	-5.1
		MCWB	-7.0	-6.6	-3.9	-0.2	6.4	12.0	15.0	14.1	10.4	2.8	-1.0	-6.2
	10%	DB	-8.2	-8.9	-3.6	1.8	9.2	16.2	19.1	17.6	12.5	4.2	-2.1	-7.8
		MCWB	-9.4	-10.2	-5.4	-0.9	5.0	10.9	14.2	13.3	9.8	1.8	-3.1	-9.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.4	-1.6	-0.6	3.1	10.7	15.5	18.6	17.5	14.1	6.7	1.7	-0.9
		MCDB	-0.2	-0.3	1.1	6.9	17.9	22.0	24.6	22.2	16.5	8.4	2.8	-0.5
	2%	WB	-3.8	-4.2	-2.3	1.5	8.4	13.9	16.8	16.0	12.5	4.7	0.4	-3.4
		MCDB	-2.2	-2.7	-0.4	4.3	14.0	20.2	22.1	19.8	14.1	6.6	1.4	-2.4
	5%	WB	-7.0	-6.6	-3.7	0.3	6.8	12.5	15.7	14.8	11.5	3.3	-1.1	-6.1
		MCDB	-5.6	-5.3	-1.8	2.6	11.2	17.6	20.4	18.0	13.0	5.2	0.0	-5.1
	10%	WB	-9.4	-10.2	-5.4	-0.7	5.3	11.3	14.5	13.9	10.5	2.1	-3.2	-8.9
		MCDB	-8.2	-9.1	-3.7	1.2	8.7	15.6	18.3	16.6	11.9	3.7	-2.3	-7.8
Mean Daily Temperature Range	5% DB	MDBR	12.5	13.3	14.3	11.6	9.6	11.1	10.1	9.5	9.9	10.4	12.0	11.3
		MCDBR	10.1	9.0	13.1	12.0	15.0	16.4	15.5	14.4	12.1	10.5	6.7	9.0
		MCWBR	9.0	8.1	10.9	8.4	8.9	9.0	8.6	8.4	7.8	7.7	5.9	8.3
	5% WB	MCDBR	9.7	8.5	11.4	10.0	13.9	14.6	13.6	11.9	9.2	9.2	6.2	9.1
		MCWBR	8.9	7.9	9.6	7.1	8.6	8.5	8.0	7.4	6.3	7.1	5.7	8.4
Clear-Sky Solar Irradiance	taub	0.229	0.275	0.334	0.428	0.461	0.428	0.483	0.437	0.349	0.316	0.258	0.214	
	taud	2.056	2.079	1.957	1.754	1.854	2.193	2.054	2.177	2.405	2.197	2.037	1.964	
	Ebn at Noon	549	714	766	744	757	803	739	746	774	652	478	399	
	Edh at Noon	45	79	124	185	180	131	148	121	81	70	45	32	
All-Sky Solar Radiation	RadAvg	0.43	1.36	3.03	4.77	5.27	5.22	4.72	3.58	2.55	1.45	0.55	0.23	
	RadStd	0.04	0.10	0.19	0.33	0.49	0.57	0.57	0.45	0.34	0.16	0.07	0.02	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	-1.61	+1.01	+0.65	+0.57	N/A	N/A	+33	N/A
Regional (1 neighbor)		N/A	N/A	-1.61	+1.01	+0.65	+0.57	N/A	N/A	+33	N/A